AEROTHON 2026 • HARDWARE DEPLOYMENT MANUAL

Real-World Implementation Guide: Raspberry Pi 5 & Cube Orange+

Complete Physical Wiring, Flight Controller Parameters, SIYI Camera Integration & Safety Protocols

Airframe: Holybro X500 Quadcopter | **Autopilot:** CubePilot Cube Orange Plus | **Companion:** Raspberry Pi 5 (4GB / 8GB) | **Optics:** SIYI Ethernet Camera (RTSP)

1. Physical Interconnections & Electrical Architecture

Deploying autonomous vision on a physical multirotor requires strict isolation between digital signal buses and high-current motor circuits to eliminate brownouts, voltage spikes, and ground loops.

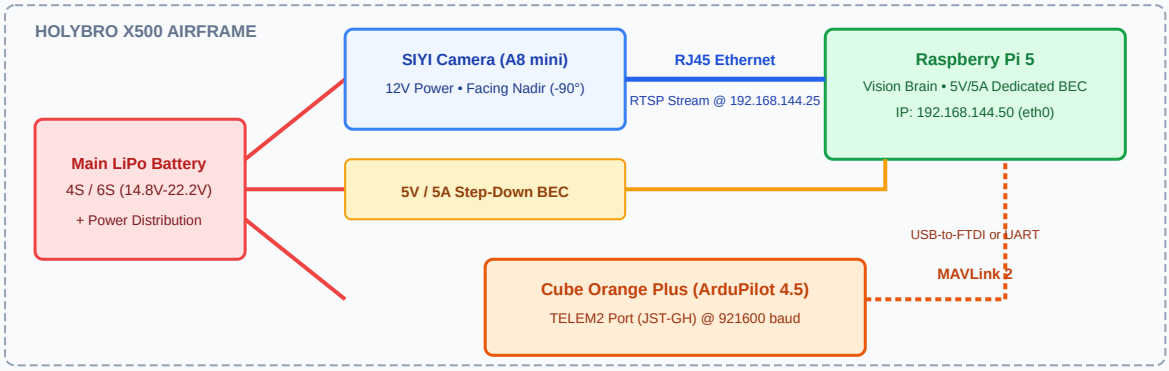


Figure 1: Complete electrical schematic and communication wiring for Holybro X500, Cube Orange+, Raspberry Pi 5, and SIYI.

CRITICAL POWER SAFETY RULES

1. **NEVER power the Raspberry Pi 5 from the Pixhawk/Cube:** The Cube's internal 5V bus is strictly rated for 1A max (for sensors and RC receivers). The Pi 5 draws up to 5A under high CPU/OpenCV load. Power the Pi 5 exclusively from an independent **5V / 5A step-down BEC** wired to the main battery.
2. **Common Ground Reference:** Ensure the ground (GND) wires of the battery, Cube Orange Plus, Raspberry Pi 5, and SIYI camera are all tied together to establish a stable reference potential for UART and Ethernet signals.
3. **SIYI Power:** The SIYI A8 mini / ZR10 requires 11V to 26V. Power it from a 12V BEC or directly from a 4S LiPo. **Do not connect it to 5V!**

2. Cube Orange Plus Parameter Configuration (Mission Planner)

Connect your Cube Orange Plus via USB to Mission Planner or QGroundControl and verify the following parameters in the Full Parameter List:

Parameter	Required Value	Technical Function & Rationale
`SERIAL2_PROTOCOL`	`2`	Enables MAVLink 2 framing on the `TELEM2` port for high-rate telemetry and setpoints.
`SERIAL2_BAUD`	`921`	Sets baud rate to **921,600 bps** to eliminate serial buffer latency during 20 Hz setpoint streaming.
`ARMING_CHECK`	`1`	Enforces all flight controller safety checks (Compass calibration, GPS 3D lock, EKF convergence).
`FS_GCS_ENABLE`	`1`	Triggers automatic Return-To-Launch (RTL) if companion computer MAVLink heartbeats cease.
`FENCE_ENABLE`	`1`	**Geofence Safety:** Enforces an invisible containment boundary to prevent flyaway.
`FENCE_RADIUS`	`35.0`	Restricts quadcopter travel within a 35-meter radius of the takeoff point.
`FENCE_ALT_MAX`	`10.0`	Caps maximum flight altitude to 10 meters above ground level.
`WPNAV_SPEED`	`150.0`	Configures survey traversal speed to 1.5 m/s (150 cm/s) for optimal QR optical capture.
`MNT1_TYPE`	`4` or `0`	Gimbal control: Lock camera pitch facing downward at -90° (Nadir).

3. Emergency Manual Takeover Setup (Mandatory Safety Rule)

When running autonomous code that commands `GUIDED` and `BRAKE` modes, the safety pilot **must retain instantaneous manual override**. Configure your RC transmitter (RadioMaster TX16S, FrSky, etc.):

Flight Mode Switch (Channel 5 - 3 Position)

- **Position 1 (Up):** `ALT_HOLD` (Manual throttle & attitude)
- **Position 2 (Mid):** `LOITER` (GPS position lock & hover)
- **Position 3 (Down):** `GUIDED` (Autonomous script control)

Emergency BRAKE Switch (Channel 7 or 8)

- Assign a dedicated toggle switch to `BRAKE` mode.
- Flipping this switch immediately freezes the drone in place, overriding companion computer commands instantly.

4. Raspberry Pi 5 Operating System & Network Configuration

1. Static IP Configuration for SIYI Ethernet Link

The SIYI camera is factory-configured at IP `192.168.144.25` on subnet `192.168.144.xxx`. The Pi 5's Ethernet port (`eth0`) must be assigned a static IP in the same subnet:

```
# Create persistent static IP connection for SIYI on Raspberry Pi 5
sudo nmcli con add type ethernet con-name siyi-eth ifname eth0 ip4 192.168.144.50/24
sudo nmcli con up siyi-eth

# Test Ethernet connection to SIYI camera
ping -c 3 192.168.144.25
```

2. Serial USB / UART Permissions

```
# Allow non-root access to Pixhawk USB/Serial port
sudo usermod -aG dialout $USER
```

3. Install Lightweight Python Dependencies

On the Raspberry Pi 5, you **do not need to install ROS 2**. Running the mission controller in pure Python with OpenCV and MAVLink uses less than 15% CPU:

```
sudo apt update
sudo apt install -y python3-pip python3-opencv libzbar0
pip3 install pymavlink pyzbar numpy
```

5. Software Integration: Connecting Your SIYI Feed

Since you already have a working Python script to capture the SIYI feed, you can integrate it directly into `lawnmower_qr_mission.py` using this lightweight multi-threaded capture class:

```
# Threaded SIYI Camera Capture Adapter (Zero Frame Latency)
import threading
import cv2

class SiyiStreamWorker:
    def __init__(self, capture_source):
        # Uses your custom SIYI capture object or direct RTSP stream
        self.cap = cv2.VideoCapture(capture_source, cv2.CAP_FFMPEG)
        self.frame = None
        self.lock = threading.Lock()
        self.running = True
        # Background worker continuously discards stale buffer frames
        threading.Thread(target=self._update, daemon=True).start()

    def _update(self):
        while self.running:
            ret, img = self.cap.read()
            if ret:
                with self.lock:
                    self.frame = img

    def get_frame(self):
        with self.lock:
            return None if self.frame is None else self.frame.copy()
```

6. Real-World Georeferencing vs. Simulation Coordinates

In the Gazebo simulation, the takeoff pad was hard-coded at `(N = -25.0, E = 0.0)` because the world origin was placed at the field center.

> **THE REAL-WORLD DIFFERENCE**
> When the Cube Orange Plus acquires a GPS 3D fix outdoors and arms, **ArduPilot EKF3 automatically initializes Local NED `(N = 0.0, E = 0.0)` at the exact physical spot where the drone took off!**

Therefore, in real life, configure your search zone **relative to your physical takeoff pad**:

- **Takeoff Pad:** Local NED `(N = 0.0m, E = 0.0m)`.
- **Search Area:** Set the lawnmower survey to sweep the field in front of the pad (e.g. North from `+5.0m` to `+30.0m`, East from `-12.0m` to `+12.0m`).

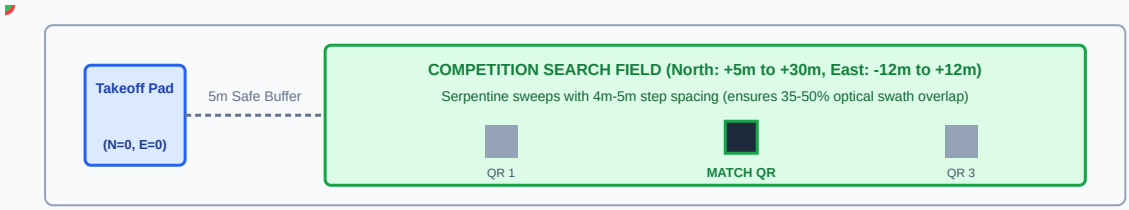


Figure 2: Real-world field layout relative to physical takeoff pad (N=0, E=0).

7. Four-Stage Zero-Risk Flight Testing Protocol

Do not attempt full autonomous flight immediately. Strictly execute this phased validation sequence:

Stage 1: Benchtop Test (PROPELLERS OFF! ⚠️)

1. Remove all four propellers from the X500 frame.
2. Power on the drone. SSH into the Raspberry Pi 5 over Wi-Fi.
3. Run the mission script in bench mode:

```
python3 scripts/lawnmower_qr_mission.py --url /dev/ttyUSB0:921600 --alt 4.0
```

4. Hold your printed reference QR under the SIYI camera. Verify console prints: `[SUCCESS] REFERENCE QR MEMORIZED INTO MEMORY!`
5. Slide the twin candidate QR under the camera. Verify console logs: `[MATCH FOUND!] 100.0%` and confirms `Got COMMAND_ACK: DO_SET_MODE: ACCEPTED (Mode BRAKE)`.

Stage 2: Manual Flight & Loiter Stability Verification

1. Mount propellers. Place the drone in an open outdoor test field.
2. Arm and takeoff manually in `ALT_HOLD` mode to 4.0 meters.
3. Switch RC to `LOITER` mode. Verify the drone holds rock-solid position with GPS lock (minimum 14 satellites, HDOP < 0.9).
4. Flip your emergency RC `BRAKE` switch. Verify the drone stops in place immediately.

Stage 3: Companion Computer Single-Waypoint Test

1. Hover the drone manually at 4.0m in `LOITER`. Flip flight mode switch to `GUIDED`.
2. On the Raspberry Pi 5, command a single 5-meter translation:

```
python3 scripts/fly_to.py --target-n 5.0 --target-e 0.0 --target-d -4.0 --url /dev/ttyUSB0:921600
```

3. Verify the drone smoothly cruises 5 meters forward and stops. Flip your RC switch back to `LOITER` to confirm manual override works instantly.

Stage 4: Full Autonomous Mission Execution

1. Place the 2m × 2m reference QR tile directly under the drone on the takeoff spot.
2. Place the twin candidate QR and distractor tiles in the search field ahead.
3. Ensure geofence is active (`FENCE_ENABLE = 1`).
4. Keep one finger on your RC transmitter mode switch (ready to flip to `LOITER` or `ALT_HOLD` at any millisecond).
5. Launch the autonomous mission script on the Raspberry Pi 5:

```
python3 scripts/lawnmower_qr_mission.py \
  --url /dev/ttyUSB0:921600 \
  --alt 4.0 \
  --speed 1.5 \
  --step 4.5 \
  --align-duration 15.0 \
  --no-camera
```

(Note: `--no-camera` disables the X11 GUI window to save CPU cycles on headless Pi).

6. **Expected Autonomous Flight Sequence:**
   - Drone arms in `GUIDED` mode and climbs to 4.0m altitude.
   - Holds steady for 1.5s, downward SIYI camera captures reference QR, extracts 21×21 module grid into RAM.
   - Transitions smoothly across the 5m buffer to entry waypoint `(N = 5.0m, E = -10.0m)`.
   - Performs lawnmower serpentine survey across the field at 1.5 m/s.
   - The moment twin QR is spotted (score ≥ 88%), issues targeted `BRAKE` command, stopping horizontal momentum.
   - Pinhole ray projection calculates ground coordinates `(N_target, E_target)`.
   - Drone flies directly to the target QR, initiates closed-loop visual servoing to lock the optical crosshair over the QR centroid with 0.7 px precision (< 1 cm error).
   - Maintains steady alignment hover directly over the QR for 15.0 seconds.

8. Pre-Flight Field Verification Checklist

#	Safety / Subsystem Item	Required Status	Sign-off
1	**Main LiPo Battery Voltage**	≥ 4.15V per cell (16.6V for 4S, 24.9V for 6S)	[ ] PASS
2	**Dedicated Pi 5 5V/5A BEC**	Supplying steady 5.1V under load; Pi PWR LED solid red (no under-voltage)	[ ] PASS
3	**SIYI Camera Gimbal Pitch**	Locked facing straight downward (-90° Nadir); lens clean & unoccluded	[ ] PASS
4	**Ethernet Connection to SIYI**	`ping -c 2 192.168.144.25` returns 0% packet loss, latency < 2ms	[ ] PASS
5	**MAVLink Link on TELEM2**	921,600 baud; heartbeat active; telemetry stream rate ≥ 20 Hz	[ ] PASS
6	**GPS Satellites & HDOP**	Satellites ≥ 14, HDOP ≤ 0.8; EKF status green	[ ] PASS
7	**Geofence Active**	Radius 35m, Altitude 10m ceiling confirmed in Mission Planner	[ ] PASS
8	**RC Transmitter Manual Override**	Mode switch (LOITER / ALT_HOLD) and emergency BRAKE switch tested & verified	[ ] PASS

READY FOR AUTONOMOUS FLIGHT
With all 8 checklist items verified, your Holybro X500 is completely qualified for safe, fully autonomous QR survey and precision visual servoing.

Safety Pilot Field Clearance

Safety Pilot: _____
Signature: _____
Date / Time: _____

Mission Systems & Software Lead

Systems Lead: _____
Signature: _____
Airframe S/N: _____